

## **CHAPTER III**

### **SYSTEM ANALYSIS**

This chapter explains about the requirements and analysis in order to achieve the objectives in the end of the thesis. This chapter includes System Overview, Functional Analysis, Software Requirements, User's Hardware and Software Requirements, Use Case Diagram, Use Case Narrative, Swim lane diagram, and Sequence Diagram.

#### **3.1. System Overviews**

The development of this thesis is focuses on creating an application that help people to control their weights through healthy ways in order to promote healthy living that in turn will sustain any changes achieved when using this application. This thesis runs on Android mobile devices which are having purposed to give the user flexibility and mobility of a mobile devices to use anywhere and whenever. The application is named "Diet Project" is according to what a "Project" means in people minds that an effort to improve "something" in a set amount of time that will have affect to a longer amount of time in the future. This thesis is aiming to improve the Life quality of the users by doing healthy lifestyle that will affecting their lives in the future.

As an application that focuses in health this application take a great deal of research in determining the limit of human bodies. This application should know a space limit that this application able to contribute. At the first time of running this application the user will be asked to input their data like Height, Weight, age,

gender, Current activity level, and their weekly target that can be chosen between losing or gaining weight. After that the application will start to count the Calorie target of the user. Calorie target is a number representation of how much you could ate if you want to reach your target.

After that the user start in the Main Activity of the application. The main activity will show the user how much the target calorie how much they eat and finally how much calorie they have workout. In the Main Activity the user also able to add activity like Breakfast, Lunch, Dinner, Snacks, and Workout. The user can add activity by tapping the correspondent button and will be move to the activity page here they can see all the activity they have been doing at that day and add more activity. If the user add more activity they can tap the button ad will be presented to a list of Food/Workout after they choose what they want to add the application will count the calorie they take/spend and show it in the main activity so that user can know the status of their target at that day. In case that user can't find the food/workout they looking for user can easily add new food and workout given they know the amount of calorie the food/workout have. To assist the user to create a healthy lifestyle for them the applications are able to help the user to remind the user in case of daily workout or food to eat.

This thesis is developed in Java. All the scripting process will be done with the default editor of Android namely Android Studio. This thesis will combine simple and clean interface, and intuitive user experience as well as features in Android touch screen that will make the user easier to control.

### 3.2.Functional Requirement

Diet Project has a several requirements. The functional requirements of the systems can be seen in Table 3.1

Table 3.1 Functional requirement table

| No. | Description of Functional Requirements                                            |
|-----|-----------------------------------------------------------------------------------|
| 1   | It is able to Count Body Mass Ratio, Daily Calorie Needs, and User Calorie Target |
| 2   | It is able to record the amount and time of food have been eaten by the users     |
| 3   | It is able to record the amount and time of Workout have been Done by the users   |
| 4   | It is able to delete food that have been Done by the users                        |
| 5   | It is able to delete Workout that have been Done by the users                     |
| 6   | It is able to Add new food and save it in database                                |
| 7   | It is able to Add new Workout and save it in database                             |

There are 7 requirement that the program must fulfill to be considered a success.

Each of these are rally at each other. Each of these Function must be full filled by DietProject application in order to give full experience to the users.

### 3.3. Software Requirement

The development of Escape to Diet Project application is supported by some software to make sure the application works and runs properly. The software that used listed as follows:

1. Operating System

This application is developed using Windows 7 Operating System.

2. Programming IDE

To do the scripting, this application is used Android Studio as the main IDE

3. Script Language

Programming language that used for this application is Java

4. Software Development Tools

Android SDK is used to as software development tools to deploy the application into Android Platform.

### 3.4. User Requirement

The hardware and software requirements to run Diet Project for the users are listed in Table 3.2. The recommended requirements to run the application properly are:

Table 3.2 Table User Requirement

| No | Specifications                                                 |
|----|----------------------------------------------------------------|
| 1  | Android Device                                                 |
| 2  | Operating System – Android Gingerbread version 2.3.3 or higher |
| 3  | Processor – 800 MHz                                            |

|   |                        |
|---|------------------------|
| 4 | Memory – 512 MB of RAM |
|---|------------------------|

### 3.5. Use Case Diagram

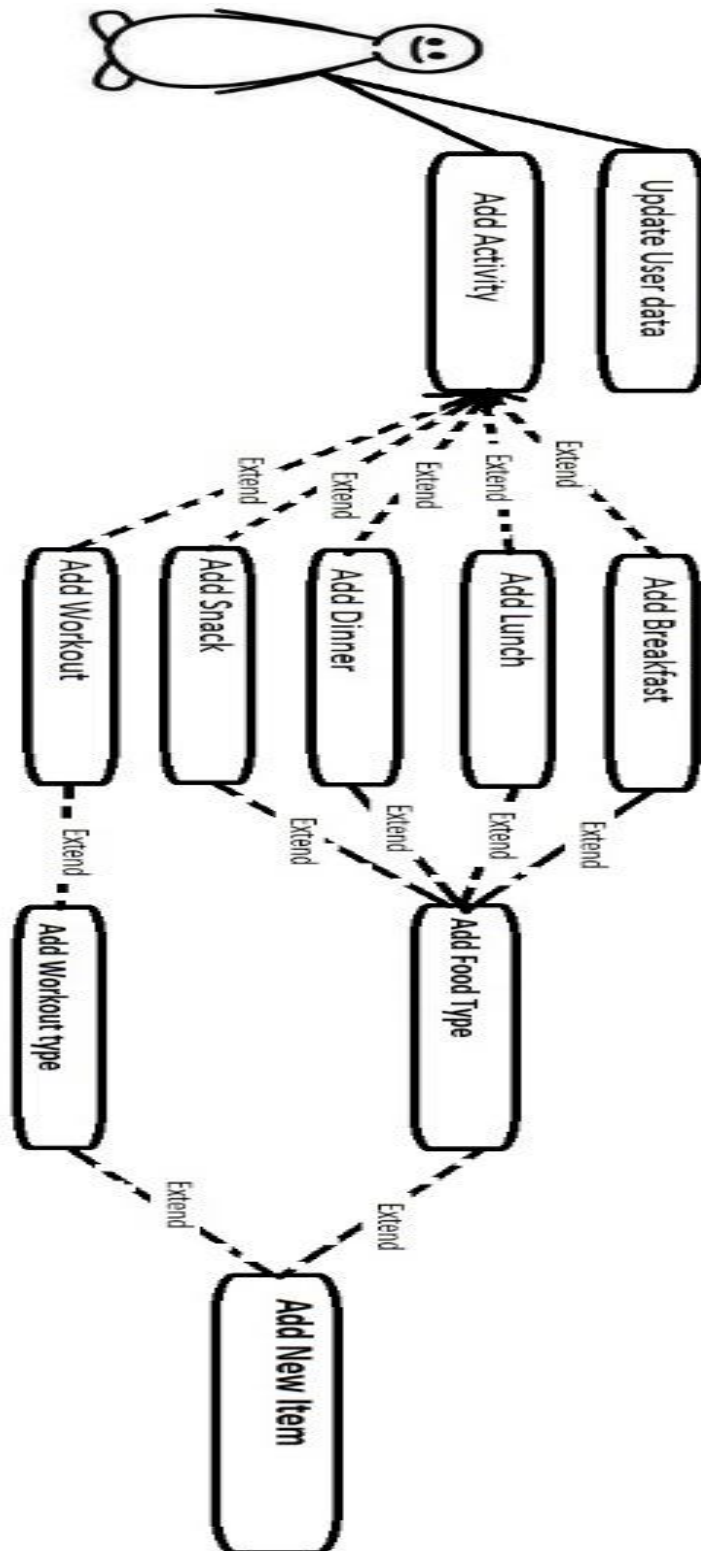


Figure 3.1 Use Case for Diet Project

A use case describes the behavior of a system. That description is written from the point of view of a user who has just told the system to do something particular. Use case diagrams have only 4 major elements: The actors who interacts with the system, the system itself, the use cases, or services, that the system knows how to perform, and the lines that represent relationships between these elements. The use case diagram of Diet Project application is shown in Figure 3.1.

For Creating Diet project there are 9 Use cases, which are: Update User data, Add Activity, Add Workout, Add Snack, Add Dinner, Add Lunch, Add Breakfast, Add Workout Type, and Add Food.

### 3.6.Use Case Narrative

The Use Case Narrative is a further description of Use Case Diagram that provides more elements and represented in a table. Use Case Narrative is also could explain the Use Case Diagram more details. Based on that definition,

Table 3.3 Update user Data

|                        |                                                                                                                                              |        |
|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|--------|
| Use Case Name          | Update User Data                                                                                                                             |        |
| Use Case Id            | UUD                                                                                                                                          |        |
| Priority               | High                                                                                                                                         |        |
| Primary Business Actor | User                                                                                                                                         |        |
| Primary System Actor   | Application                                                                                                                                  |        |
| Description            | This Use case describe user action of Updating the user's data                                                                               |        |
| Pre-Condition          | User Tapped Update Data Button                                                                                                               |        |
| Trigger                | This Use case initiated when user tapped the Update Data button and also can initiated if this is the first time User start the Application. |        |
|                        | Actor                                                                                                                                        | System |

|                          |                                                                                                                                                                                                                  |                                                                              |
|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|
| Typical Course Of Action |                                                                                                                                                                                                                  | Step 1: load User previously Saved data.                                     |
|                          | Step 2: Change the data and tap Save Button.                                                                                                                                                                     | Step 3: record data changes and send user back to Application's Home screen. |
| Alternate Course         | This Course happen when User press back button without saving                                                                                                                                                    |                                                                              |
|                          | Actor                                                                                                                                                                                                            | System                                                                       |
|                          |                                                                                                                                                                                                                  | Step 1: load User previously Saved data.                                     |
|                          | Step 2 : tap back button                                                                                                                                                                                         | Step 3 : Ignore any changes and revert to the last saved data                |
| Conclusion               | This Use case shows the step how user can easily update their data                                                                                                                                               |                                                                              |
| Post Condition           | System give the result such as :<br><br>Updated User data and updated Application home screen when player choose to save the changes      Return to Home screen without any changes when user press back button. |                                                                              |

Table 3.3 tells us about the Use case Update user data. User can update their date whenever they deem necessary as long as they press the save button then the system automatically sends user back to application home screen. If the user choose to press back button all the unsaved changes are considered forfeited and will not be saved.

Table 3.4 Add Activity Use Case

|                        |                                                           |
|------------------------|-----------------------------------------------------------|
| Use Case Name          | Add Activity                                              |
| Use Case Id            | AA                                                        |
| Priority               | High                                                      |
| Primary Business Actor | User                                                      |
| Primary System Actor   | Application                                               |
| Description            | This Use case describe user action of Adding new activity |
| Pre-Condition          | User Tapped Add Activity Button                           |

|                          |                                                                                            |                                               |
|--------------------------|--------------------------------------------------------------------------------------------|-----------------------------------------------|
| Trigger                  | This Use case initiated when user tapped the Add Activity button                           |                                               |
| Typical Course Of Action | Actor                                                                                      | System                                        |
|                          |                                                                                            | Step 1 : load Add Activity page               |
|                          | Step 2 : Choose kind of activities to add                                                  | Step 3 : show the corresponding activity page |
| Alternate Course         | This Course happen when User press back button                                             |                                               |
|                          | Actor                                                                                      | System                                        |
|                          |                                                                                            | Step 1 : load Add Activity page               |
|                          | Step 2 : tap back button                                                                   | Step 3: send User back to Main activity page. |
| Conclusion               | This Use case shows the step how user can Adding activity                                  |                                               |
| Post Condition           | User at Activity page when the user choose any activity otherwise user back at home screen |                                               |

Table 3.4 shows us how user can add activity to their database. Users can add activity that can be differentiate by 5 categories: Breakfast, Lunch, dinner, Snack, and Workout.

Table 3.5 Add Breakfast Use Case

|                        |                                                                   |
|------------------------|-------------------------------------------------------------------|
| Use Case Name          | Add Breakfast                                                     |
| Use Case Id            | AB                                                                |
| Priority               | High                                                              |
| Primary Business Actor | User                                                              |
| Primary System Actor   | Application                                                       |
| Description            | This Use case describe user action of Adding new breakfast items  |
| Pre-Condition          | User Tapped Add Breakfast Button                                  |
| Trigger                | This Use case initiated when user tapped the Add Breakfast button |

|                          |                                                                                           |                                              |
|--------------------------|-------------------------------------------------------------------------------------------|----------------------------------------------|
| Typical Course Of Action | Actor                                                                                     | System                                       |
|                          |                                                                                           | Step 1 : load Add Breakfast page             |
|                          | Step 2 : Choose Add Food button                                                           | Step 3 : show the Add Food page              |
| Alternate Course         | This Course happen when User press back button                                            |                                              |
|                          | Actor                                                                                     | System                                       |
|                          |                                                                                           | Step 1 : load Add Breakfast page             |
|                          | Step 2 : tap back button                                                                  | Step 3: send User back to Add Activity page. |
| Conclusion               | This Use case shows the step how user can Adding activity Breakfast                       |                                              |
| Post Condition           | User at Food page when the user choose Add Food otherwise user back at Add Activity page. |                                              |

Table 3.5 shows us how user add activity (Breakfast). In this use case user can add breakfast. This use case could be accessed via the add activity menu.

Table 3.6 Add Lunch Use Case.

|                          |                                                               |                              |
|--------------------------|---------------------------------------------------------------|------------------------------|
| Use Case Name            | Add Lunch                                                     |                              |
| Use Case Id              | AL                                                            |                              |
| Priority                 | High                                                          |                              |
| Primary Business Actor   | User                                                          |                              |
| Primary System Actor     | Application                                                   |                              |
| Description              | This Use case describe user action of Adding new lunch items  |                              |
| Pre-Condition            | User Tapped Add Lunch Button                                  |                              |
| Trigger                  | This Use case initiated when user tapped the Add Lunch button |                              |
| Typical Course Of Action | Actor                                                         | System                       |
|                          |                                                               | Step 1 : load Add Lunch page |

|                  |                                                                                           |                                              |
|------------------|-------------------------------------------------------------------------------------------|----------------------------------------------|
|                  | Step 2 : Choose Add Food button                                                           | Step 3 : show the Add Food page              |
| Alternate Course | This Course happen when User press back button                                            |                                              |
|                  | Actor                                                                                     | System                                       |
|                  |                                                                                           | Step 1 : load Add Lunch page                 |
|                  | Step 2 : tap back button                                                                  | Step 3: send User back to Add Activity page. |
| Conclusion       | This Use case shows the step how user can Adding activity Lunch                           |                                              |
| Post Condition   | User at Food page when the user choose Add Food otherwise user back at Add Activity page. |                                              |

Table 3.6 shows us how user add activity (Lunch). Similar with add breakfast it can be accessed via the add activity menu.

Table 3.7 Add Dinner Use Case.

|                          |                                                                |                                 |
|--------------------------|----------------------------------------------------------------|---------------------------------|
| Use Case Name            | Add Dinner                                                     |                                 |
| Use Case Id              | AD                                                             |                                 |
| Priority                 | High                                                           |                                 |
| Primary Business Actor   | User                                                           |                                 |
| Primary System Actor     | Application                                                    |                                 |
| Description              | This Use case describe user action of Adding new Dinner items  |                                 |
| Pre-Condition            | User Tapped Add Dinner Button                                  |                                 |
| Trigger                  | This Use case initiated when user tapped the Add Dinner button |                                 |
| Typical Course Of Action | Actor                                                          | System                          |
|                          |                                                                | Step 1 : load Add Dinner page   |
|                          | Step 2 : Choose Add Food button                                | Step 3 : show the Add Food page |
| Alternate Course         | This Course happen when User press back button                 |                                 |

|                |                                                                                           |                                              |
|----------------|-------------------------------------------------------------------------------------------|----------------------------------------------|
|                | Actor                                                                                     | System                                       |
|                |                                                                                           | Step 1 : load Add Dinner page                |
|                | Step 2 : tap back button                                                                  | Step 3: send User back to Add Activity page. |
| Conclusion     | This Use case shows the step how user can Adding activity Dinner                          |                                              |
| Post Condition | User at Food page when the user choose Add Food otherwise user back at Add Activity page. |                                              |

Table 3.7 shows us how user add activity (Dinner). Similar with add breakfast it can be accessed via the add activity menu.

Table 3.8 Add Snack Use Case.

|                          |                                                                 |                                              |
|--------------------------|-----------------------------------------------------------------|----------------------------------------------|
| Use Case Name            | Add Snack                                                       |                                              |
| Use Case Id              | AS                                                              |                                              |
| Priority                 | High                                                            |                                              |
| Primary Business Actor   | User                                                            |                                              |
| Primary System Actor     | Application                                                     |                                              |
| Description              | This Use case describe user action of Adding new Snack items    |                                              |
| Pre-Condition            | User Tapped Add Snack Button                                    |                                              |
| Trigger                  | This Use case initiated when user tapped the Add Snack button   |                                              |
| Typical Course Of Action | Actor                                                           | System                                       |
|                          |                                                                 | Step 1 : load Add Snack page                 |
|                          | Step 2 : Choose Add Food button                                 | Step 3 : show the Add Food page              |
| Alternate Course         | This Course happen when User press back button                  |                                              |
|                          | Actor                                                           | System                                       |
|                          |                                                                 | Step 1 : load Add Snack page                 |
|                          | Step 2 : tap back button                                        | Step 3: send User back to Add Activity page. |
| Conclusion               | This Use case shows the step how user can Adding activity Snack |                                              |

|                |                                                                                           |
|----------------|-------------------------------------------------------------------------------------------|
| Post Condition | User at Food page when the user choose Add Food otherwise user back at Add Activity page. |
|----------------|-------------------------------------------------------------------------------------------|

Table 3.8 shows us how user add activity (Snack). Similar with add breakfast it can be accessed via the add activity menu.

Table 3.9 Add Workout Use Case.

|                          |                                                                                                           |                                              |
|--------------------------|-----------------------------------------------------------------------------------------------------------|----------------------------------------------|
| Use Case Name            | Add Workout                                                                                               |                                              |
| Use Case Id              | AW                                                                                                        |                                              |
| Priority                 | High                                                                                                      |                                              |
| Primary Business Actor   | User                                                                                                      |                                              |
| Primary System Actor     | Application                                                                                               |                                              |
| Description              | This Use case describe user action of Adding new Workout items                                            |                                              |
| Pre-Condition            | User Tapped Add Workout Button                                                                            |                                              |
| Trigger                  | This Use case initiated when user tapped the Add Workout button                                           |                                              |
| Typical Course Of Action | Actor                                                                                                     | System                                       |
|                          |                                                                                                           | Step 1 : load Add Workout page               |
|                          | Step 2 : Choose Add Workout type button                                                                   | Step 3 : show the Add Workout type page      |
| Alternate Course         | This Course happen when User press back button                                                            |                                              |
|                          | Actor                                                                                                     | System                                       |
|                          |                                                                                                           | Step 1 : load Add Workout page               |
|                          | Step 2 : tap back button                                                                                  | Step 3: send User back to Add Activity page. |
| Conclusion               | This Use case shows the step how user can Adding activity Workout                                         |                                              |
| Post Condition           | User at Workout type page when the user choose Add Workout type otherwise user back at Add Activity page. |                                              |

Table 3.9 shows us how user add activity (Workout). Similar with add breakfast it can be accessed via the add activity menu.

Table 3.10 Add Workout type Use Case.

|                          |                                                                                            |                                                                                          |
|--------------------------|--------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| Use Case Name            | Add Workout Type                                                                           |                                                                                          |
| Use Case Id              | AWT                                                                                        |                                                                                          |
| Priority                 | High                                                                                       |                                                                                          |
| Primary Business Actor   | User                                                                                       |                                                                                          |
| Primary System Actor     | Application                                                                                |                                                                                          |
| Description              | This Use case describe user action of Adding or choosing new Workout items                 |                                                                                          |
| Pre-Condition            | User Tapped Add Workout Type Button                                                        |                                                                                          |
| Trigger                  | This Use case initiated when user tapped the Add Workout Type button                       |                                                                                          |
| Typical Course Of Action | Actor                                                                                      | System                                                                                   |
|                          |                                                                                            | Step 1 : load Add Workout Type page                                                      |
|                          | Step 2 : Choose Workout type button                                                        | Step 3 : show the Add Workout page and send selected workout to be stored by Add Workout |
| Alternate Course         | This Course happen when User press New item button                                         |                                                                                          |
|                          | Actor                                                                                      | System                                                                                   |
|                          |                                                                                            | Step 1 : load Add Workout type page                                                      |
|                          | Step 2 : tap add new item and fill the new item                                            | Step 3: send User back to Add Workout type page.                                         |
| Conclusion               | This Use case shows the step how user can choose Workout                                   |                                                                                          |
| Post Condition           | User at Workout page when the user choose A Workout type item otherwise user add new item. |                                                                                          |

Add Workout type is used when the user wanted to add new item to their daily workout. This can be accessed from the add workout menu. User can choose the kind of workout they want to do and the added workout will be appeared at the add workout page.

Table 3.11 Add Food Use Case.

|                          |                                                                                            |                                                                                                                      |
|--------------------------|--------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| Use Case Name            | Add Food                                                                                   |                                                                                                                      |
| Use Case Id              | AF                                                                                         |                                                                                                                      |
| Priority                 | High                                                                                       |                                                                                                                      |
| Primary Business Actor   | User                                                                                       |                                                                                                                      |
| Primary System Actor     | Application                                                                                |                                                                                                                      |
| Description              | This Use case describe user action of Adding or choosing new Food items                    |                                                                                                                      |
| Pre-Condition            | User Tapped Add Food Button                                                                |                                                                                                                      |
| Trigger                  | This Use case initiated when user tapped the Add Workout Type button                       |                                                                                                                      |
| Typical Course Of Action | Actor                                                                                      | System                                                                                                               |
|                          |                                                                                            | Step 1 : load Add Food page                                                                                          |
|                          | Step 2 : Choose a food item                                                                | Step 3 : show the Add breakfast/lunch/dinner/snack page and send selected food to the corresponding page to be saved |
| Alternate Course         | This Course happen when User press New item button                                         |                                                                                                                      |
|                          | Actor                                                                                      | System                                                                                                               |
|                          |                                                                                            | Step 1 : load Add food page                                                                                          |
|                          | Step 2 : tap add new item and fill the new item                                            | Step 3: send User back to Add Workout type page.                                                                     |
| Conclusion               | This Use case shows the step how user can choose Workout                                   |                                                                                                                      |
| Post Condition           | User at Workout page when the user choose A Workout type item otherwise user add new item. |                                                                                                                      |

Table 3.11 shows us how user add Food. User can add food to their daily activity record to know how much calorie they have eaten in a day. User can simply tap the desired food then it will ask how much portion the user have taken and give and save the result in the activity screen.

Table 3.12 Add Items Use Case.

|                          |                                                                                            |                                                                   |
|--------------------------|--------------------------------------------------------------------------------------------|-------------------------------------------------------------------|
| Use Case Name            | Add Item                                                                                   |                                                                   |
| Use Case Id              | AI                                                                                         |                                                                   |
| Priority                 | High                                                                                       |                                                                   |
| Primary Business Actor   | User                                                                                       |                                                                   |
| Primary System Actor     | Application                                                                                |                                                                   |
| Description              | This Use case describe user action of Adding new items                                     |                                                                   |
| Pre-Condition            | User Tapped Add Items Button                                                               |                                                                   |
| Trigger                  | This Use case initiated when user tapped the Add Items button                              |                                                                   |
| Typical Course Of Action | Actor                                                                                      | System                                                            |
|                          |                                                                                            | Step 1 : System load Add Food page                                |
|                          | Step 2 :User Choose Add Items Button                                                       | Step 3 : show the Add Items page and Ask user about the new items |
|                          | Step 4 : add the desired items data and press add                                          | Step 4 : saves the data                                           |
| Alternate Course         | This Course happen when User press New item button                                         |                                                                   |
|                          | Actor                                                                                      | System                                                            |
|                          |                                                                                            | Step 1 : load Add food page                                       |
|                          | Step 2 : tap add new item and fill the new item                                            | Step 3: send User back to Add Workout type page.                  |
| Conclusion               | This Use case shows the step how user can choose Workout                                   |                                                                   |
| Post Condition           | User at Workout page when the user choose A Workout type item otherwise user add new item. |                                                                   |

In case of the user can't find the user can manually add their wanted food or workout item to the list to be saved.

### 3.7.Swim lane diagram

The swim lane diagram is a sequence activities that are described in the flowchart form. Swimlane diagram provide richer information rather than use case diagram and usually swimlane diagram id divided into groups. This groups is divided based on who is responsible for the different steps within a process. There are three swimlane diagram in *Diet Project*.

The Swim lane diagram in Figure 3.2 shows for when user run the application.

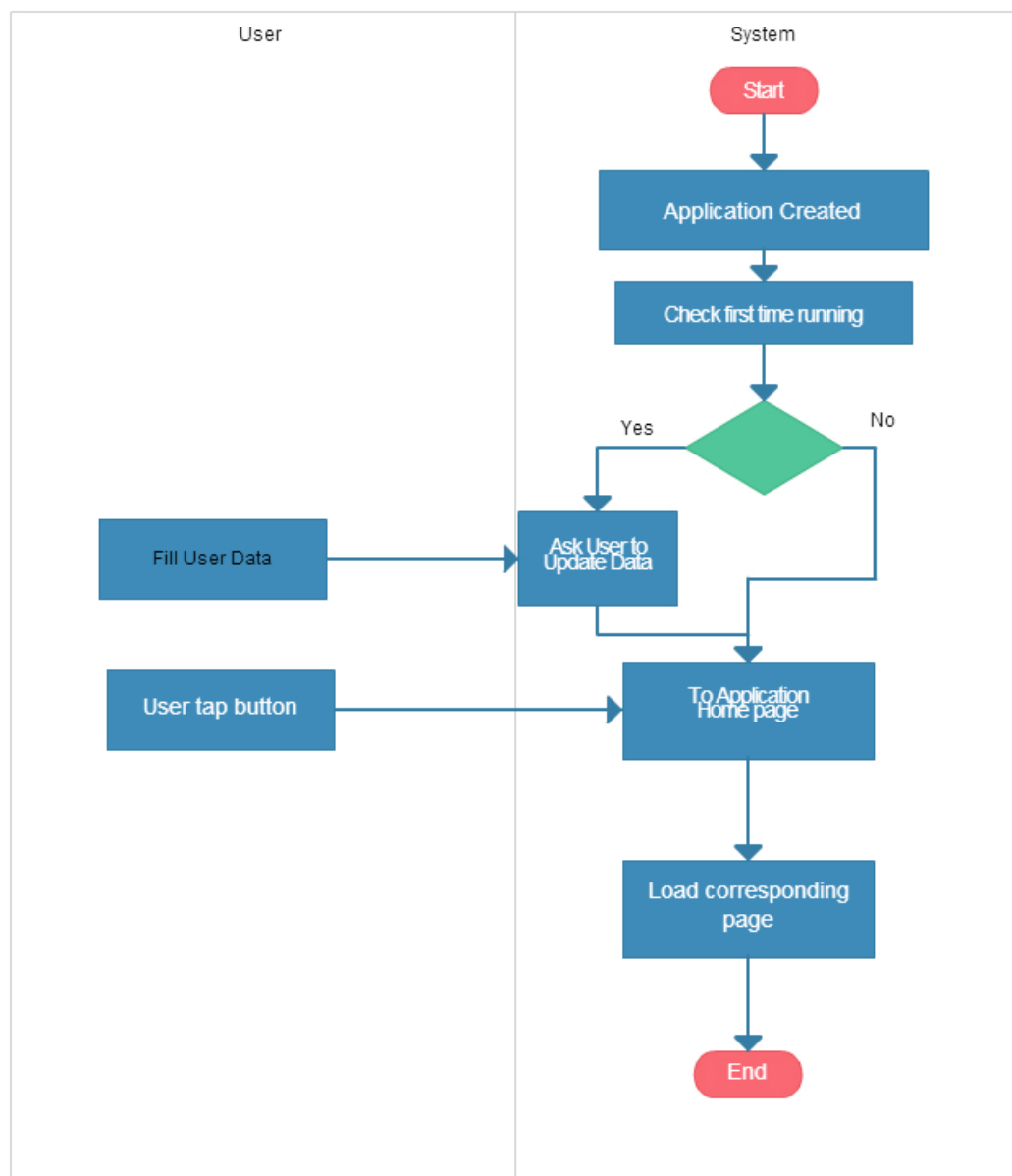


Figure 3.2 Main Activity swim lane diagram

As soon as the process is created it will first check if the application is run for the first time. If it is then it will start the User data process to collect User data for first time. Else it goes to the Main screen

The Swim lane diagram in Figure 3.3 shows for when user Update Data.

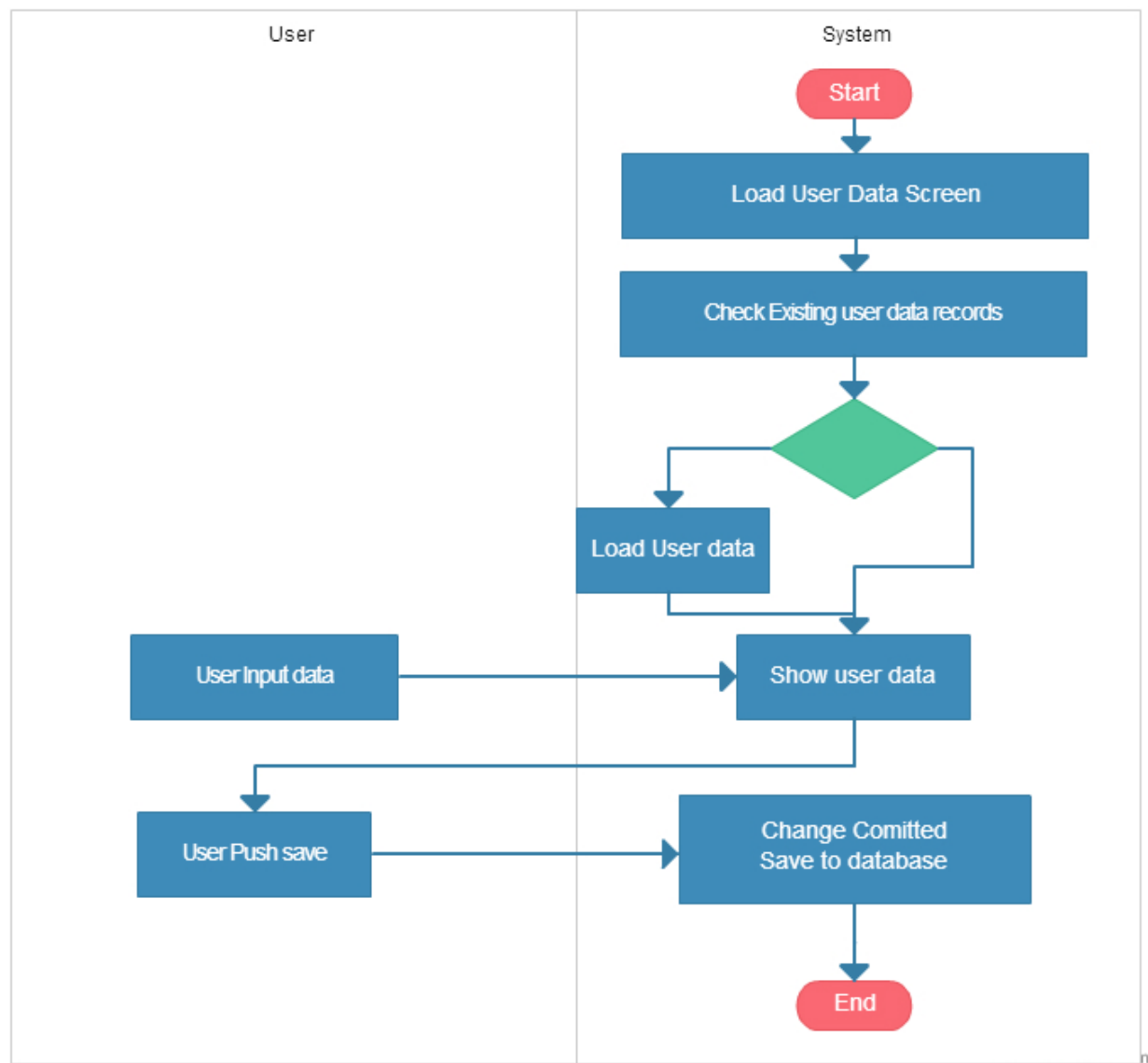


Figure 3.3 User data Swim lane diagram

As soon as the page load it will automatically check for existing data if there is data available then it will be written to the User data page else the space will remain empty.

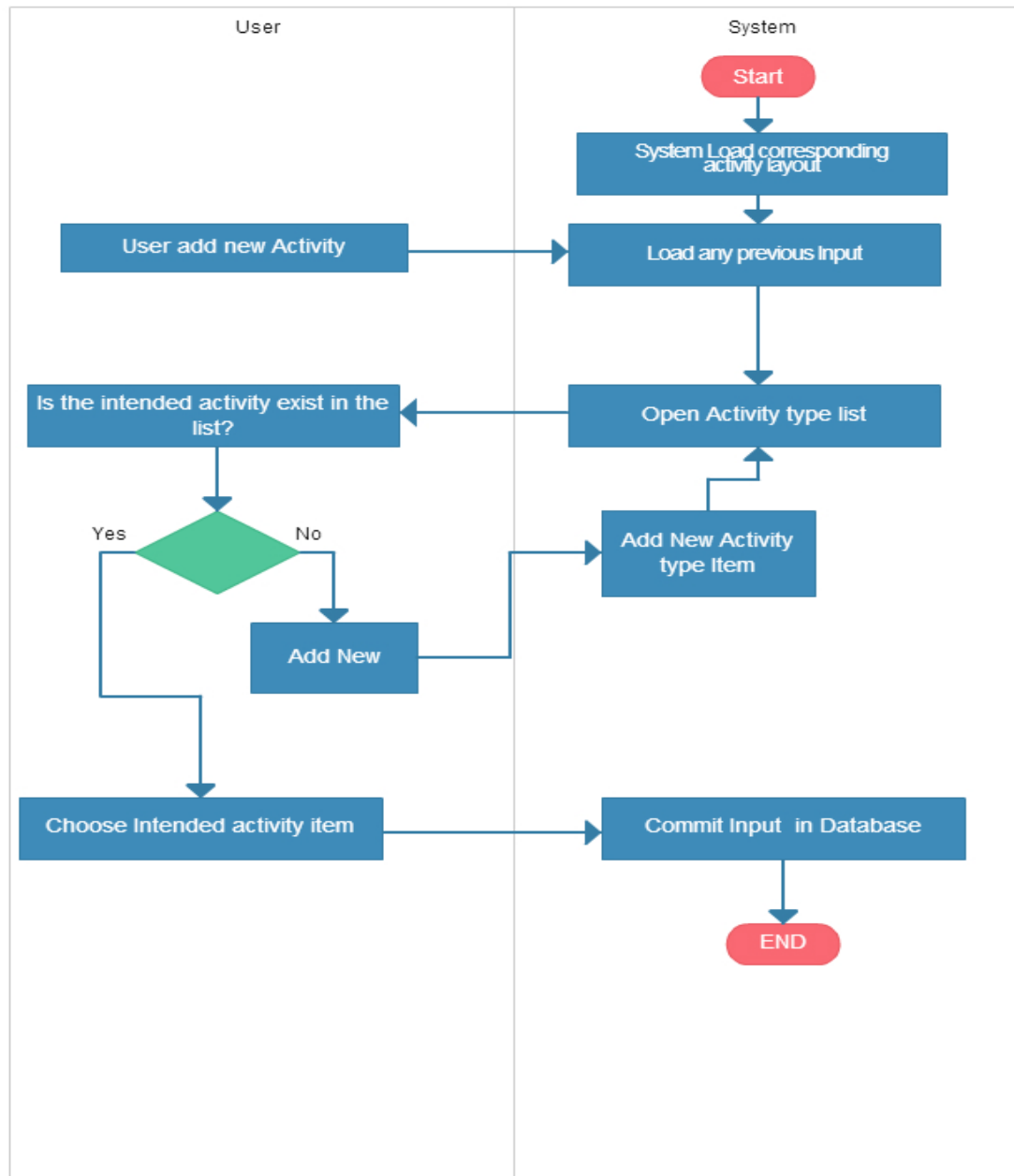


Figure 3.5

[online diagramming & design] [creately.com](https://creately.com)

Figure 3.4 Add activity Swim lane diagram

The swim lane in diagram 3.4 shows when the user add activity.

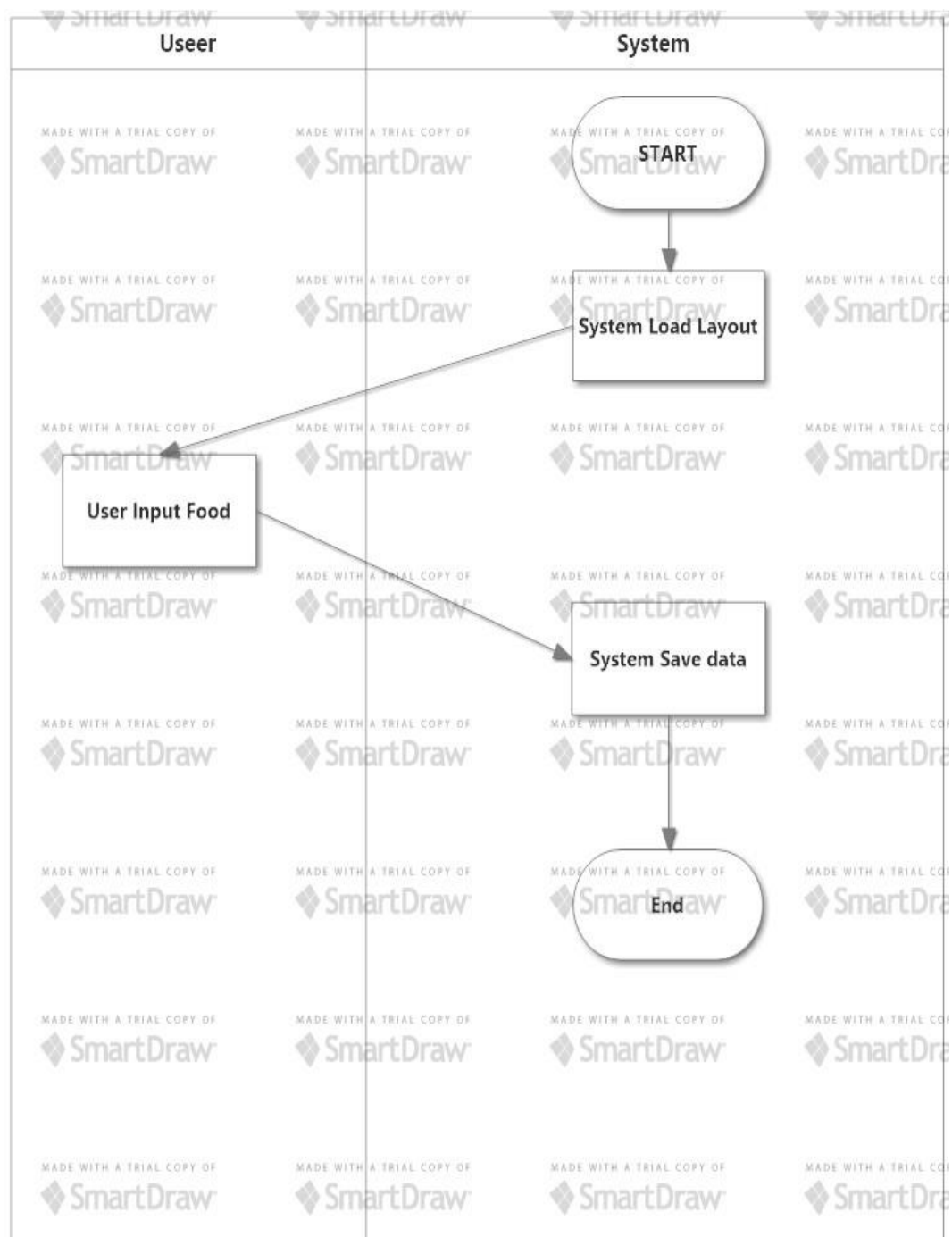


Figure 3.5 Add Breakfast Swimlane

Figure 3.5 shows us when user add Breakfast Items to Diet Project



Figure 3.6 Add Lunch Swimlane

Figure 3.6 shows us when user add Lunch Items to Diet Project

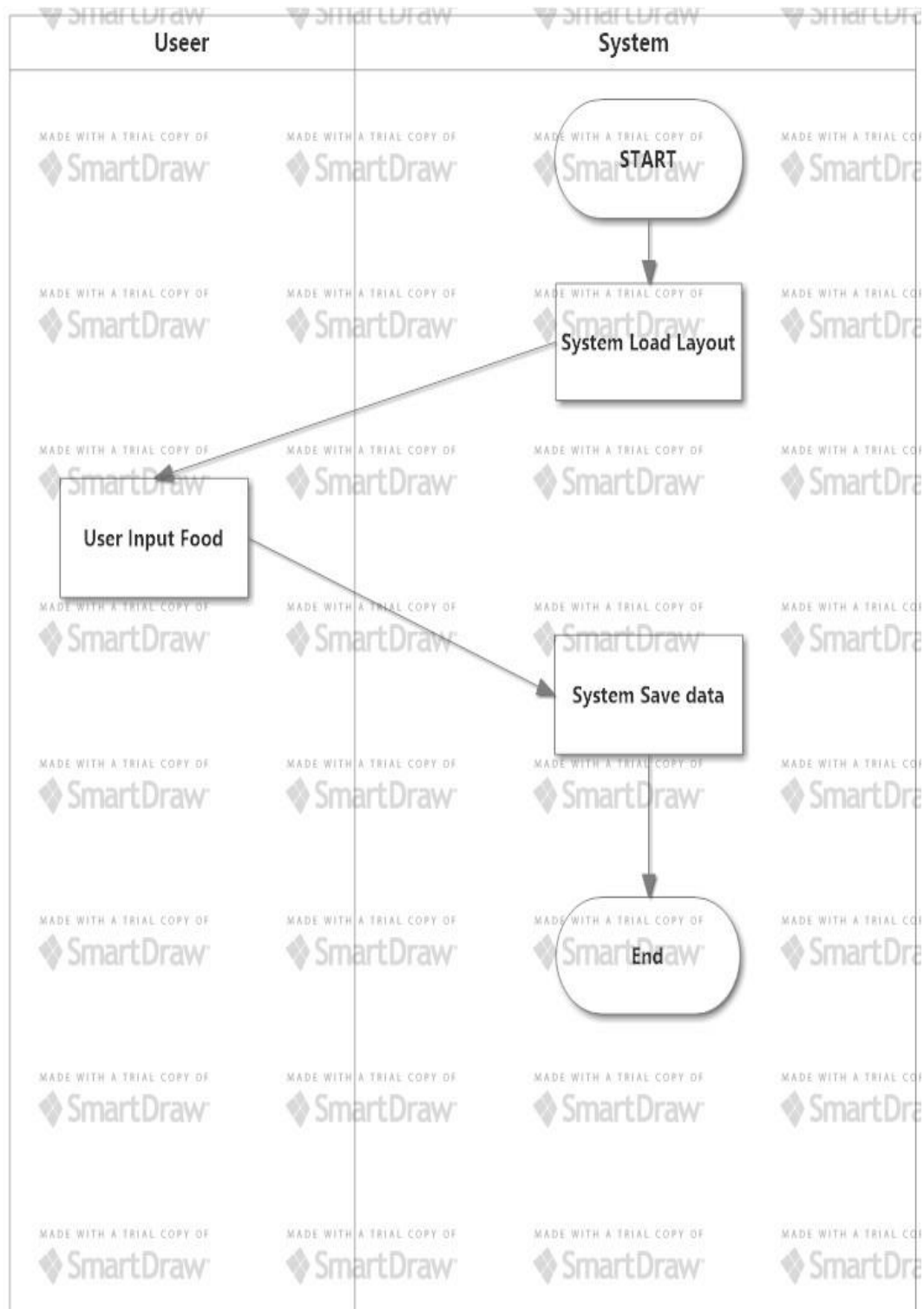


Figure 3.7 Add Dinner Swimlane

Figure 3.7 shows us when user add Dinner Items to Diet Project

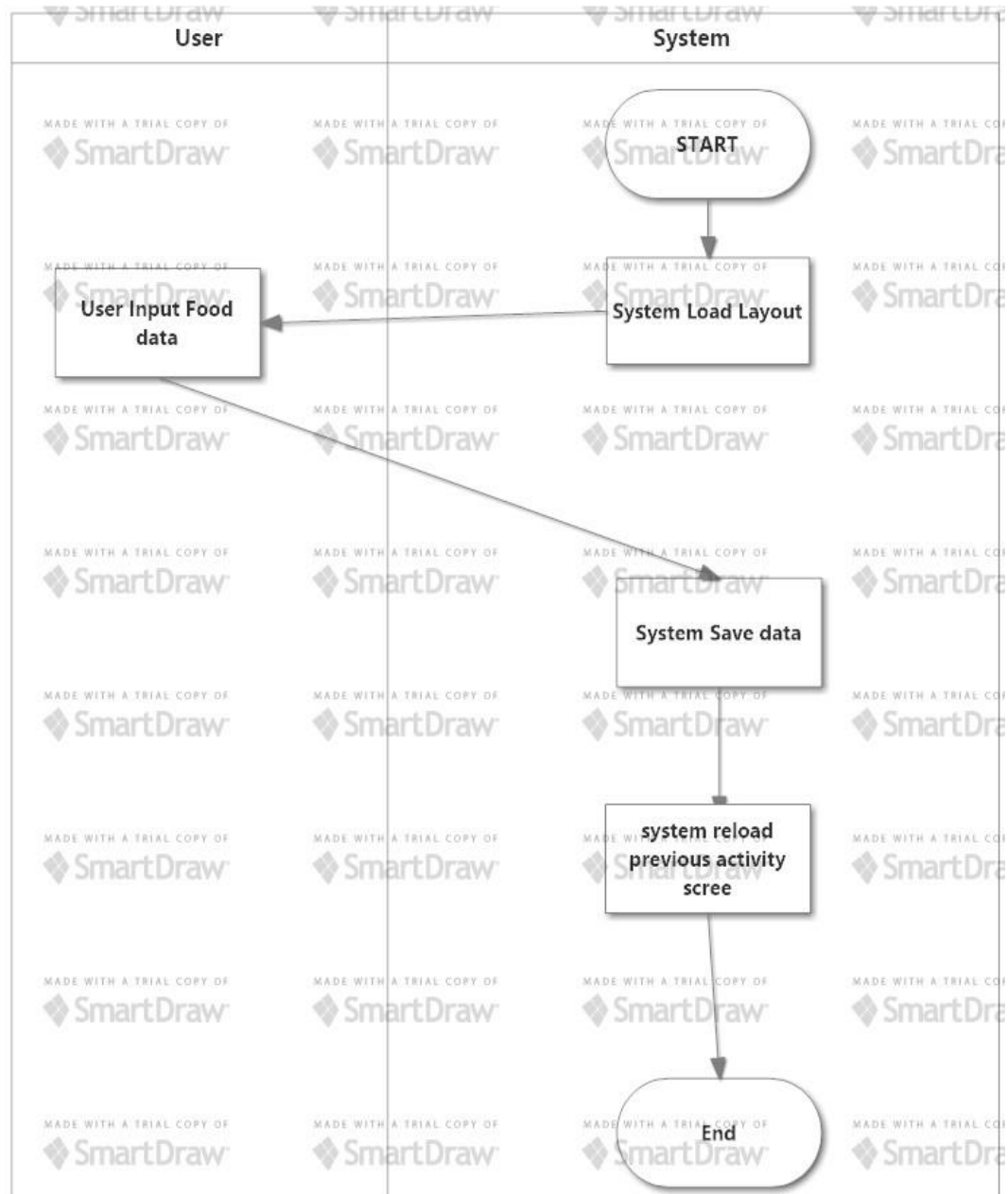


Figure 3.8 Add Food Swimlane

Figure 3.8 shows us when user add Food Items to Diet Project Database. User can easily adding new food items to the add food Swimlane.



Figure 3.9 add Workout Items Swimlane

Figure 3.9 shows us when user add Workout Items to Diet Project Database



Figure 3.10 Add Snack Swimlane

Figure 3.10 shows us when user add Snack Items to Diet Project



Figure 3.11 Add Workout Swimlane

Figure 3.11 shows us when user add Workout Items to Diet Project

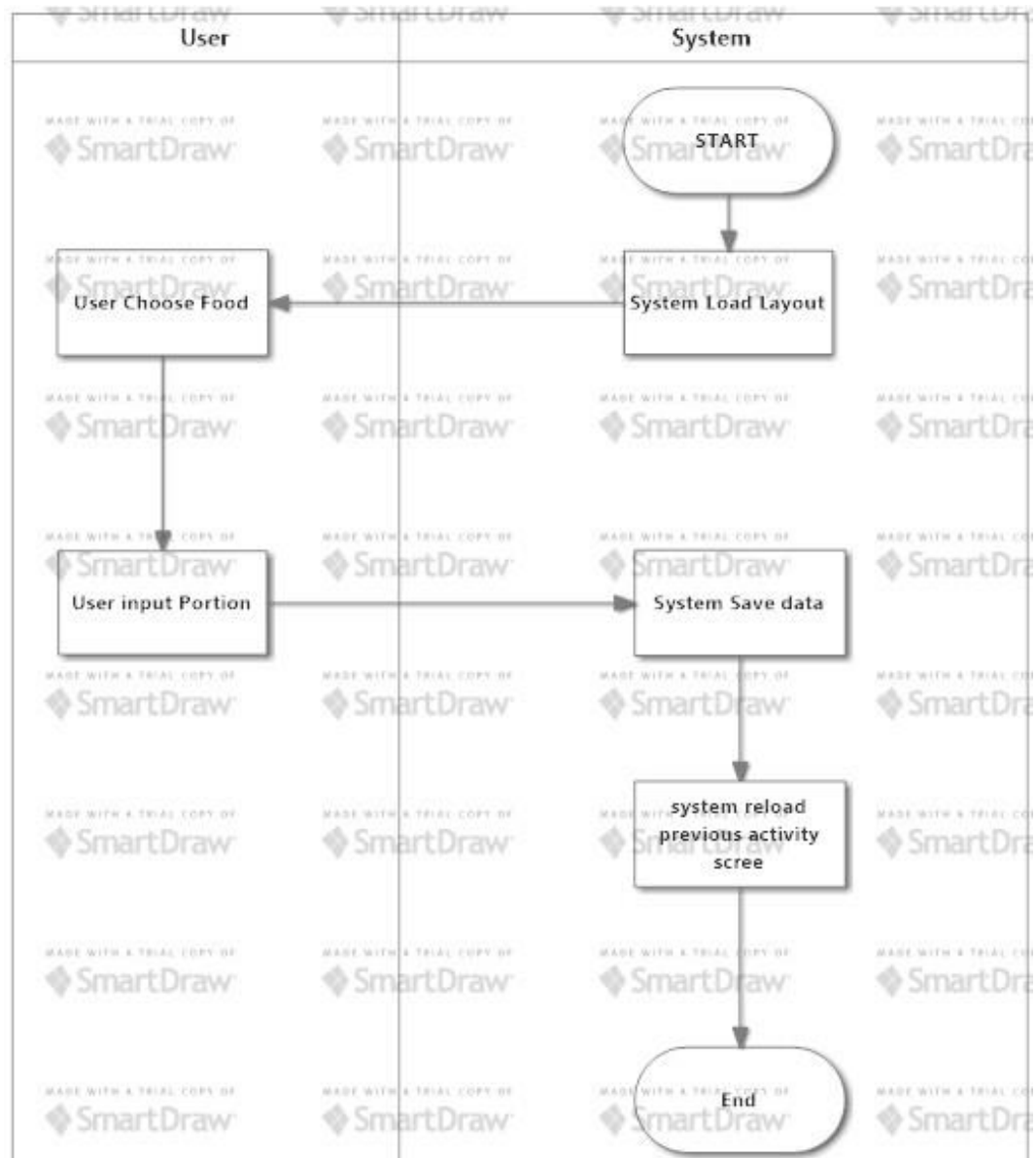


Figure 3.12 Add Food Type Swimlane

Figure 3.12 shows us when user add Food Type Items to Diet Project

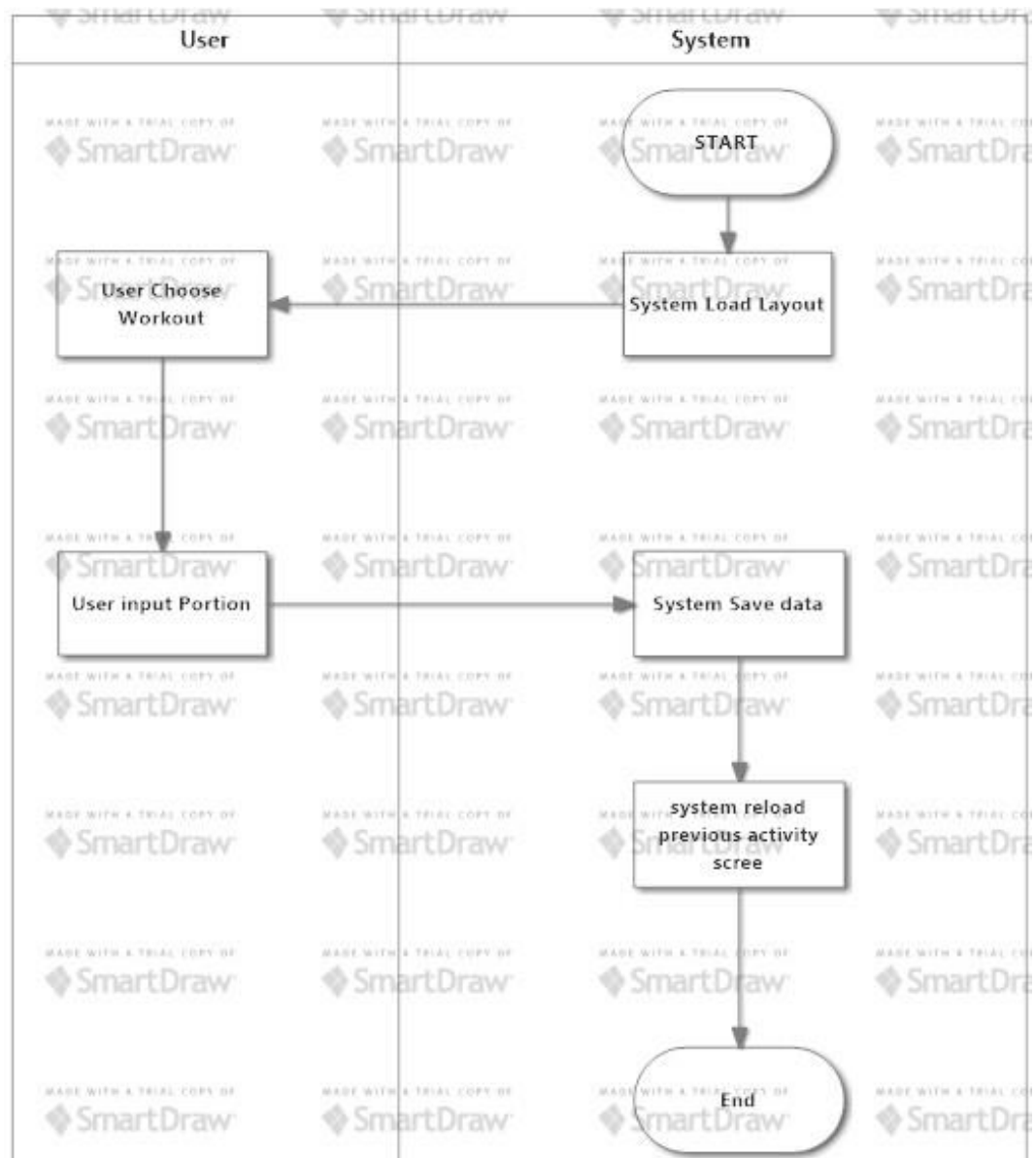


Figure 3.13 Add Workout type Swimlane

Figure 3.13 shows us when user add Workout Type Items to Diet Project